ACADEMIC PROJECT PROPOSAL

Smart Storage Monitor System

Department of Computer Science & Engineering | Intake 53

1. EXECUTIVE SUMMARY

The **Smart Storage Monitor System** is an automated, real-time environmental and security monitoring solution engineered for sensitive storage environments. By integrating temperature, humidity, motion, and acoustic sensors with an Arduino Uno microcontroller, the system continuously tracks ambient conditions and security breaches. It visually outputs live data to a local display, indicates system states via a color-coded LED matrix, and triggers an active acoustic alarm during anomaly thresholds. This low-cost, scalable system mitigates the risk of inventory spoilage, environmental degradation, and unauthorized intrusions in warehouses, archives, labs, and server rooms.

2. PROJECT TEAM & COURSE DETAILS

Name	Student ID	Intake
Al Adid	20245103007	53
Md Fahim Islam	20245103035	53
Md. Mahfuzul Bashar Shaikot	20245103038	53
Md. Sakibur Rahman	20245103091	53
Abu Ahmad Mahfuzor Rahman	20245103092	53

3. PROJECT OBJECTIVES

- **Real-time Environmental Tracking:** Continuously sample ambient temperature and relative humidity to protect perishable/sensitive inventory.
- **Security Intrusion Detection:** Monitor spatial movement and ambient noise anomalies to detect unauthorized entry or physical events.
- **Localized Diagnostics & Visualization:** Provide clear, immediate visual readouts on an I2C-enabled local display alongside a color-coded LED diagnostic panel.

- **Active Alerting System:** Implement an audible warning mechanism to immediately flag critical threshold violations (e.g., extreme heat, water leaks, or intrusion).

4. HARDWARE SYSTEM SPECIFICATIONS

Component	Quantity	Key Specifications / Purpose
Arduino Uno R3	1	ATmega328P-based microcontroller, 5V operation.
DHT22 Temp & Humidity Sensor	1	High-accuracy relative humidity and temperature sensor.
PIR Motion Sensor (HC-SR501)	1	Pyroelectric infrared sensor; adjustable range and delay.
Sound Sensor (KY-037/KY-038)	1	Microphone sensor; detects ambient decibel changes.
0.96" OLED Display / 16x2 I2C LCD	1	Local alphanumeric output interface using I2C bus.
Active Buzzer	1	High-pitch audible alert triggered on thresholds.
LED Status Indicators	5	GREEN System OK RED Temp Alert BLUE Hum Alert YELLOW Motion Alert PURPLE Sound Alert
Peripherals & Cables	1 Set	Breadboard, jumper wires, USB cable, & power supply.

5. SYSTEM ARCHITECTURE & DESIGN

Block Diagram

```
        +----------------+
        |   DHT22 Sensor |
        +--------+-------+
                 |
                 |
+-------------+  |  +----------------+
| PIR Sensor  +--+--+                |
+-------------+     |                |
                    |  Arduino Uno   |
+-------------+     |                |
|Sound Sensor +-----+                |
+-------------+     +--------+-------+
                             |
            +----------------+----------------+
            |                |                |
      +-----+----+      +-----+----+     +-----+----+
      | OLED/LCD |      |  LEDs    |     | Buzzer   |
      +----------+      +----------+     +----------+
```

Working Principle & Logic Sequence

Step 1: Sensor Sampling

The Arduino Uno continuously polls the digital input pin connected to the DHT22 and the digital output pins connected to the PIR and Sound sensors at a sample frequency of 0.5 Hz.

Step 2: Processing & Threshold Evaluation

The microcontroller parses raw environmental data and tests it against safety thresholds (e.g., Temp > 35°C, Humidity > 75%). Simultaneously, it listens for high level interrupts from security modules.

Step 3: Visual Reporting

The OLED or I2C LCD updates dynamically, showing temperature in °C, humidity in %, and localized alert alerts if flags are raised.

Step 4: Indicator & Sound Alerts

If safe thresholds are breached, the corresponding warning LED lights up immediately and the buzzer sounds a pulsed alarm. When everything is normal, only the green status LED glows.

6. PROJECTED APPLICATIONS & BENEFITS

Target Environments

- **Warehouses:** Multi-tiered food & dry goods storage.
- **Pharmaceuticals:** Cleanrooms & strict temperature cold chains.
- **Archives & Museums:** Preventing paper decay and mold growth.
- **Data Centers:** Tracking physical security and HVAC failure.

Core System Benefits

- **Early Spoilage Mitigation:** Catches temperature spikes before damage occurs.
- **Hybrid Surveillance:** Acts as both environment monitor and entry alarm.
- **Highly Cost-Efficient:** Engineered using standard open-source parts.
- **Low Operational Complexity:** Standalone hardware requires zero network overhead.